Functional Annotation Report: paaZ (Q88HT3 / PP_3270) in *Pseudomonas putida* KT2440

Summary

paaZ (UniProt **Q88HT3**; ordered locus **PP_3270**) of *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) encodes a **soluble, cytoplasmic, bifunctional enzyme** that catalyzes the pivotal ring-cleavage step of the **aerobic phenylacetate (Paa) degradation pathway**. The protein is a natural **fusion of two catalytic domains**: a C-terminal (R)-specific enoyl-CoA hydratase domain (of the MaoC/HotDog fold) that hydrolytically opens the seven-membered **oxepin-CoA** ring (**EC 3.3.2.12**, oxepin-CoA hydrolase), and an N-terminal **$NADP^+$-dependent aldehyde dehydrogenase** domain (of the Ald_DH/histidinol_DH fold) that immediately oxidizes the resulting reactive aldehyde to **3-oxo-5,6-dehydrosuberyl-CoA** (**EC 1.2.1.91**). These two activities and their domain assignments match the UniProt description and InterPro domain complement precisely, confirming that the gene symbol, organism, and domain architecture are all internally consistent — **this is the correct gene**, and it is unambiguous.

The PaaZ reaction sits at the **branch point** between an unstable, chemically reactive intermediate and productive catabolism. The oxepin-CoA ring, generated upstream by a multicomponent oxygenase (PaaABC(D)E) that epoxidizes phenylacetyl-CoA plus an isomerase that converts the epoxide, is opened by PaaZ to yield a highly reactive aldehyde. If this aldehyde is not oxidized instantly it cyclizes into a stable dead-end product; the physical fusion of the hydratase and dehydrogenase activities in a single polypeptide — and the enzyme's higher-order **trimer-of-dimers** architecture with **substrate channeling** — ensures the aldehyde is captured and oxidized before it can escape. The 3-oxo-5,6-dehydrosuberyl-CoA product is then processed by β-oxidation-like reactions, ultimately yielding **acetyl-CoA and succinyl-CoA**, which feed central metabolism.

Biologically, *paaZ* is embedded in the *paa* catabolic gene cluster, is induced in response to the true inducer **phenylacetyl-CoA** (via the PaaX repressor), and enables *P. putida* to grow on phenylacetate and on the many aromatic compounds (phenylalanine, 2-phenylethylamine,

styrene, ethylbenzene) that funnel through phenylacetyl-CoA. This same machinery has been co-opted biotechnologically for medium-chain dicarboxylate (e.g., adipate) metabolism, underscoring the pathway's chemical logic and plasticity.

Gene/Protein Identity Verification

Attribute	Value	Consistent with literature?
UniProt accession	Q88HT3	✔
Gene symbol	*paaZ*	✔ (assigned to the ring-cleavage step across taxa)
Locus	PP_3270	✔
Organism	*P. putida* KT2440	✔ (pathway characterized in *P. putida* and *E. coli*)
EC numbers	3.3.2.12 + 1.2.1.91	✔ (both directly demonstrated biochemically)
Domains	Ald_DH N/C (IPR016161/2/3, IPR015590) + HotDog/MaoC (IPR029069)	✔ (dehydrogenase + hydratase fusion)

Verification result: CONFIRMED and unambiguous. All evidence converges on a single, well-characterized bifunctional enzyme of the aerobic phenylacetate pathway. There is no symbol clash with an unrelated gene.

Key Findings

Finding 1 — PaaZ is a bifunctional oxepin-CoA hydrolase / aldehyde dehydrogenase

PaaZ is not a single-activity enzyme but a **fusion protein** combining two chemically distinct catalytic modules on one polypeptide chain. Biochemical characterization by Teufel and colleagues established that the **C-terminal (R)-specific enoyl-CoA hydratase domain** — historically annotated as "MaoC" — performs the ring-opening chemistry: it hydrolytically cleaves the oxepin-CoA ring to generate a **highly reactive aldehyde** intermediate. The **N-terminal $NADP^+$-dependent aldehyde dehydrogenase domain** then oxidizes that aldehyde to **3-oxo-5,6-dehydrosuberyl-CoA** (PMID: 21296885).

These two activities correspond exactly to the two EC numbers annotated in UniProt for Q88HT3: **EC 3.3.2.12** (oxepin-CoA hydrolase) and **EC 1.2.1.91** (3-oxo-5,6-dehydrosuberyl-CoA semialdehyde dehydrogenase). The domain assignments are also fully consistent with the InterPro complement listed in the target record — the aldehyde dehydrogenase N- and C-terminal domains (IPR016162, IPR016163, IPR015590), the shared aldehyde/histidinol DH superfamily fold (IPR016161), and the HotDog domain superfamily (IPR029069) that houses the MaoC-like hydratase. As the authors state verbatim: *"The enzyme consists of a C-terminal (R)-specific enoyl-CoA hydratase domain (formerly MaoC) that cleaves the ring and produces a highly reactive aldehyde and an N-terminal NADP(+)-dependent aldehyde dehydrogenase domain that oxidizes the aldehyde to 3-oxo-5,6-dehydrosuberyl-CoA"* (PMID: 21296885).

This finding establishes the **primary molecular function** of the gene product: an enzyme catalyzing two sequential reactions — hydrolytic ring opening followed by $NADP^+$-dependent aldehyde oxidation — with the CoA-thioester substrate providing the anchoring handle for both active sites. The **$NADP^+$** (rather than NAD^+) dependence distinguishes the fused PaaZ arrangement from the alternative seen in some bacteria that use a stand-alone NAD^+-dependent aldehyde dehydrogenase plus a separate hydratase.

Finding 2 — PaaZ acts at the ring-cleavage branch point of the aerobic phenylacetate pathway

The broader pathway context was elucidated by Teufel et al. (2010), who defined a fundamentally **novel aerobic catabolic route** distinct from classical ring-hydroxylating dioxygenase pathways. In this route, phenylacetate is first activated to **phenylacetyl-CoA**; the aromatic ring is then epoxidized to a **ring 1,2-epoxide** by a distinct multicomponent oxygenase (PaaABC(D)E); the reactive epoxide is isomerized to a **seven-membered O-heterocyclic enol ether (an oxepin)**; and this oxepin undergoes **hydrolytic ring cleavage and β-oxidation**, ultimately producing **acetyl-CoA and succinyl-CoA** (PMID: 20660314). PaaZ performs the hydrolytic ring-cleavage-plus-oxidation node of this sequence. In the authors' words: *"the aromatic ring of phenylacetyl-CoA becomes activated to a ring 1,2-epoxide by a distinct multicomponent oxygenase. The reactive nonaromatic epoxide is isomerized to a seven-member O-heterocyclic enol ether, an oxepin. This isomerization is followed by hydrolytic ring cleavage and beta-oxidation steps, leading to acetyl-CoA and succinyl-CoA"* (PMID: 20660314).

Crucially, PaaZ sits at a **metabolic branch point** because the aldehyde it generates is chemically labile. Teufel et al. (2011) demonstrated that *"If not oxidized immediately, the reactive aldehyde condenses intramolecularly to a stable cyclic derivative that is largely prevented by PaaZ fusion in vivo"* (PMID: 21296885). This provides the **physiological rationale** for why the two activities are fused into one protein: the fusion minimizes escape of the reactive intermediate into a dead-end cyclic product — a chemistry that, in other bacteria, is exploited to make tropone/tropolone natural products. PaaZ therefore functions not merely as a catalyst but as a **committed channeling device** that steers a reactive intermediate toward productive catabolism.

The downstream fate of the PaaZ product in *P. putida* was defined by Nogales et al. (2007), who characterized the final steps and confirmed that *"succinyl-CoA is one of the final products of this pathway in Pseudomonas putida and Escherichia coli"* (PMID: 17259607). Thus the 3-oxo-5,6-dehydrosuberyl-CoA generated by PaaZ is processed through β-oxidation-like steps to acetyl-CoA and succinyl-CoA, which enter the TCA cycle.

Finding 3 — PaaZ forms a trilobed trimer-of-dimers and channels its intermediate by electrostatic pivoting

The structural basis for PaaZ's channeling function was resolved by cryo-electron microscopy. Sathyanarayanan et al. (2019) determined structures of PaaZ with and without ligands and showed that **three domain-swapped dimers assemble into a trilobed hexamer** (a trimer of dimers): *"The structures reveal that three domain-swapped dimers of the enzyme form a trilobed structure"* (PMID: 31511507). This higher-order architecture spatially organizes the hydratase and dehydrogenase active sites within each functional unit.

Integrating the structures with small-angle X-ray scattering (SAXS), computation, site-directed mutagenesis, and microbial growth assays, the authors proposed a striking **substrate-channeling mechanism**. Rather than a conventional enclosed hydrophobic tunnel, PaaZ transfers its labile intermediate between the two active sites by **electrostatic pivoting of the**

CoA moiety: *"the key intermediate is transferred from one active site to the other by a mechanism of electrostatic pivoting of the CoA moiety, mediated by a set of conserved positively charged residues"* (PMID: 31511507). The CoA arm acts as a molecular tether that swings the reactive aldehyde from the ring-opening (hydratase) site to the oxidation (dehydrogenase) site, guided by conserved basic residues. This mechanism directly explains, at atomic resolution, the physiological observation from Finding 2 — the fusion and channeling together suppress formation of the dead-end cyclic product.

Finding 4 — *paaZ* is encoded in the *paa* catabolic cluster, induced by phenylacetyl-CoA, and physiologically enables aromatic catabolism

The genetic and regulatory context of *paaZ* was established in *E. coli* by Ferrández et al. (1998) and extends to *P. putida* by homology and pathway conservation. In *E. coli* the *paa* cluster is organized as *paaZ*, *paaABCDEFGHIJK*, and *paaXY*; *paaZ* is transcribed from its own promoter (Pz), and PaaZ *"appears to catalyze the third enzymatic step"* of the aerobic hybrid pathway (PMID: 9748275). Regulation is exerted by the **PaaX repressor**, which responds to the CoA-thioester intermediate rather than to free phenylacetate: *"As PA-CoA is the true inducer, PaaX becomes the first regulator of an aromatic catabolic pathway that responds to a CoA derivative"* (PMID: 9748275). This means *paaZ* expression is switched on precisely when the substrate for the pathway (phenylacetyl-CoA) accumulates. The homologous cluster and organization occur in *P. putida* and other proteobacteria such as *Azoarcus evansii* (PMID: 12189419).

Physiologically, the *paa* pathway is a **funneling hub**: diverse aromatic compounds — phenylalanine, 2-phenylethylamine, styrene, and ethylbenzene — are catabolized to the common intermediate phenylacetyl-CoA, which the pathway (including PaaZ) then degrades. In *P. putida* this supports growth on phenylacetate as a sole carbon and energy source and even underlies **chemotaxis toward phenylacetate** (mediated by the energy-taxis receptor Aer2 in *P. putida* F1) (PMID: 23377939). The same *paa*/β-oxidation machinery was more recently **co-opted for medium-chain dicarboxylate (adipate) metabolism** in engineered *P. putida* KT2440, where the *paa* cluster plays a central role in enabling growth on adipate, suberate, and sebacate (PMID: 33965615). This highlights both the native role and the biotechnological versatility of the pathway in which *paaZ* participates.

Mechanistic Model / Interpretation

PaaZ is best understood as a **two-active-site molecular assembly line** that converts a chemically dangerous intermediate into a stable, catabolizable thioester. The following diagram places PaaZ within the aerobic phenylacetate (Paa) pathway:

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Aromatic funnel substrates
(phenylalanine, 2-phenylethylamine,
 styrene, ethylbenzene, phenylacetate)
              │
              ▼
     Phenylacetyl-CoA  ◄─── PaaX / PA-CoA regulation (true inducer)
              │
              │  PaaABC(D)E  (multicomponent ring 1,2-epoxidase/oxygenase)
              ▼
     1,2-epoxyphenylacetyl-CoA
              │
              │  PaaG  (isomerase)
              ▼
        Oxepin-CoA  (7-membered O-heterocyclic enol ether)
              │
     ┌────────┴───────────────────────────────────────┐
     │                    P a a Z   (fusion)           │
     │                                                 │
     │  C-terminal MaoC/HotDog (R)-hydratase           │
     │   → hydrolytic ring opening (EC 3.3.2.12)       │
     │          │  reactive aldehyde                   │
     │          │  ↳ electrostatic pivoting of CoA moiety │
     │          ▼  (conserved + charged residues; no tunnel) │
     │  N-terminal NADP+-dependent aldehyde dehydrogenase │
     │   → oxidation (EC 1.2.1.91)                     │
     └────────┬───────────────────────────────────────┘
              ▼
   3-oxo-5,6-dehydrosuberyl-CoA
              │
              │  β-oxidation-like steps (PaaF/PaaG/PaaH/PaaJ)
              ▼
    Acetyl-CoA  +  Succinyl-CoA  →  TCA cycle

      (branch: if aldehyde escapes → stable cyclic dead-end
       product → tropone/tropolone natural products in some taxa)
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The central chemical problem the pathway must solve is that the oxepin-CoA ring, once opened, yields an aldehyde that will spontaneously cyclize into a stable, non-metabolizable dead end. Evolution's solution is elegant on three nested levels:

1. **Domain fusion** — placing the ring-opening hydratase and the aldehyde-oxidizing dehydrogenase on a single polypeptide keeps the two catalytic events tightly coupled in time and space (Finding 1, PMID: 21296885).
2. **Quaternary organization** — assembling three domain-swapped dimers into a trilobed hexamer creates the geometric framework in which the two active sites of a functional unit are correctly juxtaposed (Finding 3, PMID: 31511507).
3. **Electrostatic channeling** — using the CoA arm as a charged lever, the enzyme swings the reactive aldehyde from the hydratase site to the dehydrogenase site guided by conserved positive residues, rather than relying on a sealed hydrophobic tunnel (Finding 3, PMID: 31511507).

Together these features suppress the dead-end cyclic derivative in vivo and channel carbon into β-oxidation. The table below summarizes the two reactions and their attributes.

Attribute	Ring-opening activity	Oxidation activity
Domain	C-terminal MaoC / HotDog (R)-hydratase	N-terminal Ald_DH (aldehyde dehydrogenase)
EC number	3.3.2.12 (oxepin-CoA hydrolase)	1.2.1.91
Substrate	Oxepin-CoA	Reactive ring-opened aldehyde
Product	Reactive aldehyde (ring-opened)	3-oxo-5,6-dehydrosuberyl-CoA
Cofactor	None (hydrolytic)	$NADP^+$
InterPro fold	HotDog domain superfamily (IPR029069)	Ald_DH/histidinol_DH (IPR016161/2/3, IPR015590)

Localization. No secretion signal, membrane-spanning region, or lipid anchor is implied by the domain architecture; all substrates are soluble CoA thioesters and the cofactor is soluble $NADP^+$. Consistent with the entire Paa pathway operating on cytoplasmic CoA-activated intermediates, PaaZ carries out its function in the **cytoplasm** as a soluble oligomeric enzyme.

Pathway role summary. PaaZ catalyzes the committed conversion that transforms a ring-opened but still reactive species into a stable β-oxidation substrate. It is the linchpin connecting the oxygenase/isomerase "upper" segment of the pathway (ring activation and oxepin formation) to the "lower" segment (β-oxidation to acetyl-CoA and succinyl-CoA).

Evidence Base

PMID	Title (abbrev.)	Contribution to this report
21296885	*Studies on the mechanism of ring hydrolysis in phenylacetate degradation: a metabolic branching point*	Primary biochemical characterization defining PaaZ's two catalytic domains, the two reactions (EC 3.3.2.12 and EC 1.2.1.91), the product 3-oxo-5,6-dehydrosuberyl-CoA, and the rationale for the fusion (preventing a dead-end cyclic aldehyde). Basis for Findings 1 and 2.
20660314	*Bacterial phenylalanine and phenylacetate catabolic pathway revealed*	Landmark elucidation of the full aerobic phenylacetate pathway (epoxidation → oxepin → hydrolytic ring cleavage → β-oxidation → acetyl-CoA + succinyl-CoA) in *E. coli* and *P. putida*, placing PaaZ's step in context. Basis for Finding 2.
31511507	*Molecular basis for metabolite channeling in a ring opening enzyme of the phenylacetate degradation pathway*	Cryo-EM/SAXS/mutagenesis study revealing the trilobed trimer-of-dimers architecture and the electrostatic-pivoting channeling mechanism. Basis for Finding 3.
17259607	*Characterization of the last step of the aerobic phenylacetic acid degradation pathway*	Establishes succinyl-CoA as a final product of the pathway in *P. putida* and *E. coli*, defining the downstream fate of PaaZ's product. Supports Finding 2.
9748275	*Catabolism of phenylacetic acid in E. coli: a new aerobic hybrid pathway*	Defines the *paa* gene cluster organization, assigns PaaZ to the third enzymatic step, and establishes PaaX/phenylacetyl-CoA regulation. Basis for Finding 4.
23377939	*Taxis of P. putida F1 toward phenylacetic acid (Aer2)*	Demonstrates the physiological importance of PAA catabolism in *P. putida*, including chemotaxis via energy taxis. Supports Finding 4.
33965615	*Engineering adipic acid metabolism in P. putida*	Shows the *paa* cluster's central role when the pathway is co-opted for medium-chain dicarboxylate (adipate) metabolism. Supports Finding 4.
22582071	*Epoxy Coenzyme A Thioester pathways for degradation of aromatic compounds*	Review contextualizing the epoxide/oxepin CoA-thioester strategy of aerobic aromatic degradation

PMID	Title (abbrev.)	Contribution to this report
		(benzoate and phenylacetate). Supports the pathway framework.
12189419	*Aerobic metabolism of phenylacetic acids in Azoarcus evansii*	Independent characterization of a conserved *paa* cluster (paaYZGHIKABCDE) and pathway in another proteobacterium, supporting the generality of the pathway.
28153386	*Phenylacetaldehyde dehydrogenase from P. putida S12 styrene pathway*	Related aldehyde dehydrogenase in an upstream funneling route (styrene → phenylacetate), illustrating how aromatics feed into phenylacetyl-CoA. Contextual.
12906358	*Design of catabolic cassettes for styrene biodegradation*	Demonstrates that styrene degradation funnels into phenylacetate/PAA metabolism, supporting the funneling role of the pathway. Contextual.

Consistency with UniProt/InterPro annotation. The literature-derived functions align exactly with the target record: the two EC numbers (3.3.2.12 and 1.2.1.91), the bifunctional "oxepin-CoA hydrolase / 3-oxo-5,6-dehydrosuberyl-CoA semialdehyde dehydrogenase" description, the gene name *paaZ*, the locus PP_3270, and the InterPro domains (aldehyde dehydrogenase N/C domains and the HotDog superfamily that houses the MaoC hydratase). All verification criteria are satisfied.

Supported and Refuted Hypotheses

Supported: - PaaZ (Q88HT3/PP_3270) is a bifunctional oxepin-CoA hydrolase (EC 3.3.2.12) + $NADP^+$-dependent aldehyde dehydrogenase (EC 1.2.1.91). ✔ (PMID: 21296885) - Its substrate is the pathway-specific oxepin-CoA; its product is 3-oxo-5,6-dehydrosuberyl-CoA. ✔ (PMID: 21296885, PMID: 20660314) - It performs the ring-cleavage (third) step of the aerobic *paa* pathway, upstream of β-oxidation to succinyl-CoA/acetyl-CoA. ✔ (PMID: 9748275, PMID: 17259607) - It is a cytoplasmic trimer-of-dimers that channels its intermediate via electrostatic pivoting. ✔ (PMID: 31511507) - The fusion prevents accumulation of a toxic cyclic dead-end product. ✔ (PMID: 21296885)

Refuted / ruled out: - PaaZ is **not** an oxygenolytic ring-cleavage enzyme; ring opening is **hydrolytic** (non-oxygenolytic). (PMID: 20660314, PMID: 21296885) - The gene symbol is **not** ambiguous here; all evidence converges on a single, well-defined enzyme.

Limitations and Knowledge Gaps

1. **Direct enzymology on the *P. putida* KT2440 ortholog is limited.** The definitive biochemical (Teufel et al.) and structural (Sathyanarayanan et al.) work was performed largely on PaaZ orthologs (notably from *E. coli* and related systems). Function is assigned to Q88HT3 by strong sequence/domain homology, identical EC annotations, and conserved pathway organization rather than by direct kinetic characterization of the KT2440 protein itself. Strain-specific kinetic parameters (kcat, Km for oxepin-CoA) were not located.
2. **Cofactor specificity nuance.** The dehydrogenase domain is $NADP^+$-dependent; the strictness of $NADP^+$ vs. NAD^+ preference and its regulatory implications for the KT2440 enzyme are not experimentally quantified here.
3. **Localization is inferred, not measured.** The cytoplasmic assignment follows logically from the soluble CoA-thioester substrates and the absence of targeting/membrane domains, but no direct fractionation or imaging evidence for Q88HT3 was examined.
4. **Regulatory details in *P. putida*.** PaaX/phenylacetyl-CoA regulation is best characterized in *E. coli*. The precise promoter architecture and operon boundaries around PP_3270 in KT2440 were not directly verified.
5. **Channeling residue conservation.** The electrostatic-pivoting mechanism was demonstrated for a specific ortholog; residue-level validation in the KT2440 sequence would confirm identical implementation.

Proposed Follow-up Experiments / Actions

1. **Direct biochemical characterization of Q88HT3.** Heterologously express and purify the KT2440 PaaZ; measure steady-state kinetics for oxepin-CoA hydrolysis and aldehyde oxidation, confirm the $NADP^+$ preference, and verify the 3-oxo-5,6-dehydrosuberyl-CoA product by LC-MS.

2. **Genetic knockout / complementation in KT2440.** Delete PP_3270 and test loss of growth on phenylacetate (and on funneling substrates such as phenylalanine, 2-phenylethylamine, styrene); complement in trans. Assess accumulation of the predicted dead-end cyclic aldehyde derivative in the mutant by metabolomics.
3. **Localization confirmation.** Use cell fractionation and/or a fluorescent fusion to confirm cytoplasmic localization of PP_3270.
4. **Structural validation.** Solve or predict (AlphaFold/cryo-EM) the KT2440 PaaZ structure, confirm the trilobed trimer-of-dimers assembly, and test key conserved basic residues implicated in electrostatic CoA pivoting via site-directed mutagenesis and channeling assays.
5. **Regulatory mapping.** Define the *paaZ* (PP_3270) promoter and PaaX binding site in KT2440; confirm induction by phenylacetyl-CoA using reporter fusions.
6. **Applied metabolic engineering.** Given the demonstrated co-option of the *paa* cluster for adipate/dicarboxylate metabolism, systematically test PaaZ variants for expanded substrate scope toward plastic-monomer dicarboxylates, informing bio-upcycling strain design.

Report generated for target gene paaZ (UniProt Q88HT3, locus PP_3270) in Pseudomonas putida *KT2440. All functional claims are attributed to the primary and review literature cited above; annotation-database attributes (EC numbers, InterPro domains) were cross-checked against and found consistent with the experimental literature.*
